Less is More: Local Intrinsic Dimensions of Contextual Language Models

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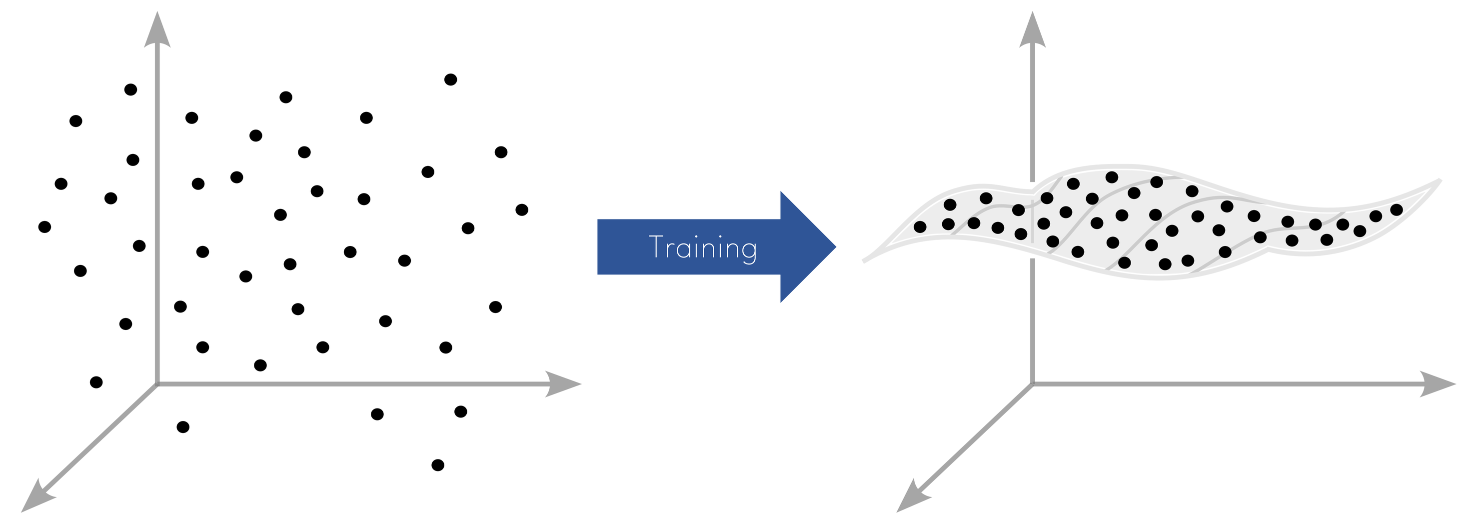
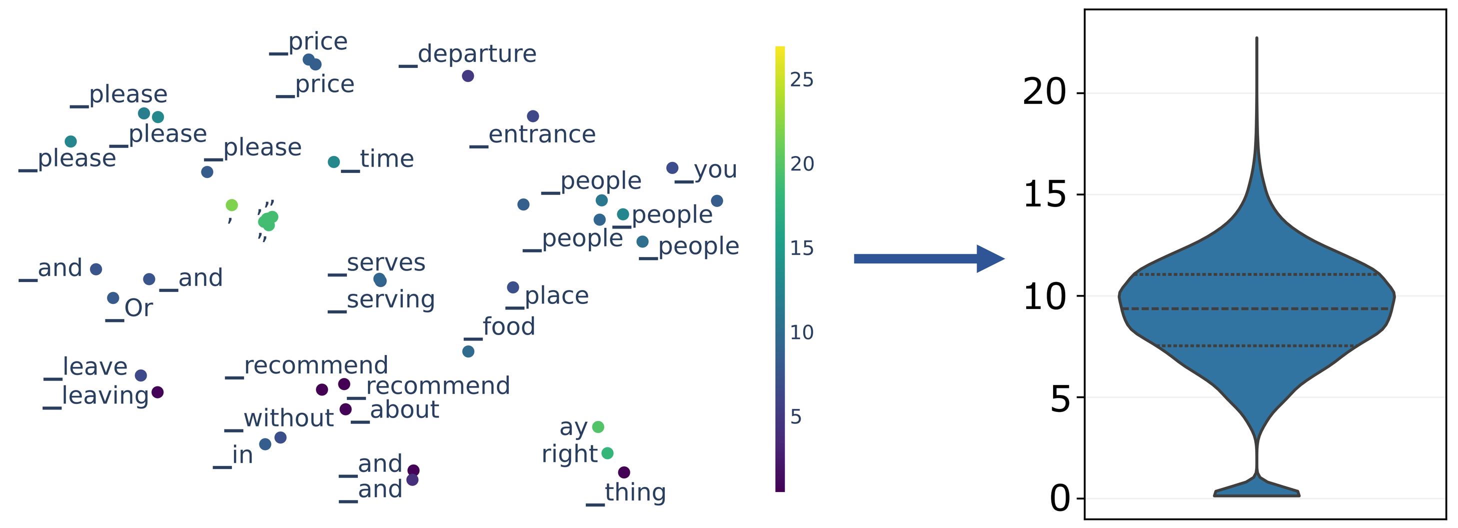
Does Embedding Geometry Reveal LLM Learning Dynamics?

- Fundamental questions, such as how fine-tuning affects model behavior, often **require extensive empirical evaluation.**
- We introduce a novel perspective based on the **geometric properties** of **contextual latent embeddings** to study the effects of training and fine-tuning.
- **Local intrinsic dimensions** provide insights into the model's training dynamics and generalization ability.

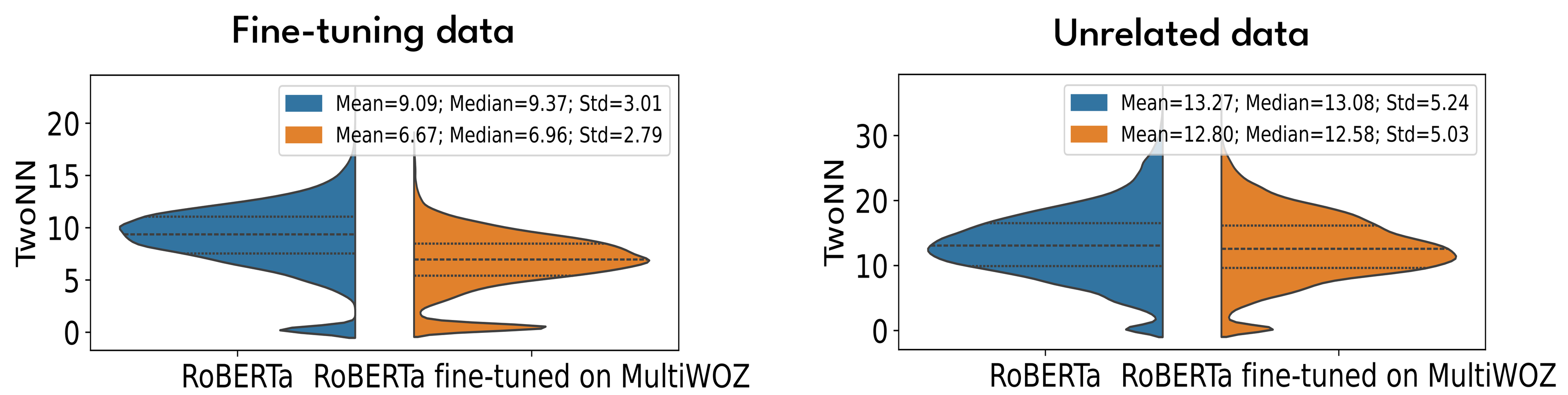


We apply a **localized TwoNN dimension estimator** to a **subsample** of contextual token embeddings, and analyse dimension **shifts** during training and fine-tuning.



Mean local dimension is stable estimate over different data subsamples and splits.

(1) Fine-Tuning Induces Dataset-Specific Dimensional Shifts

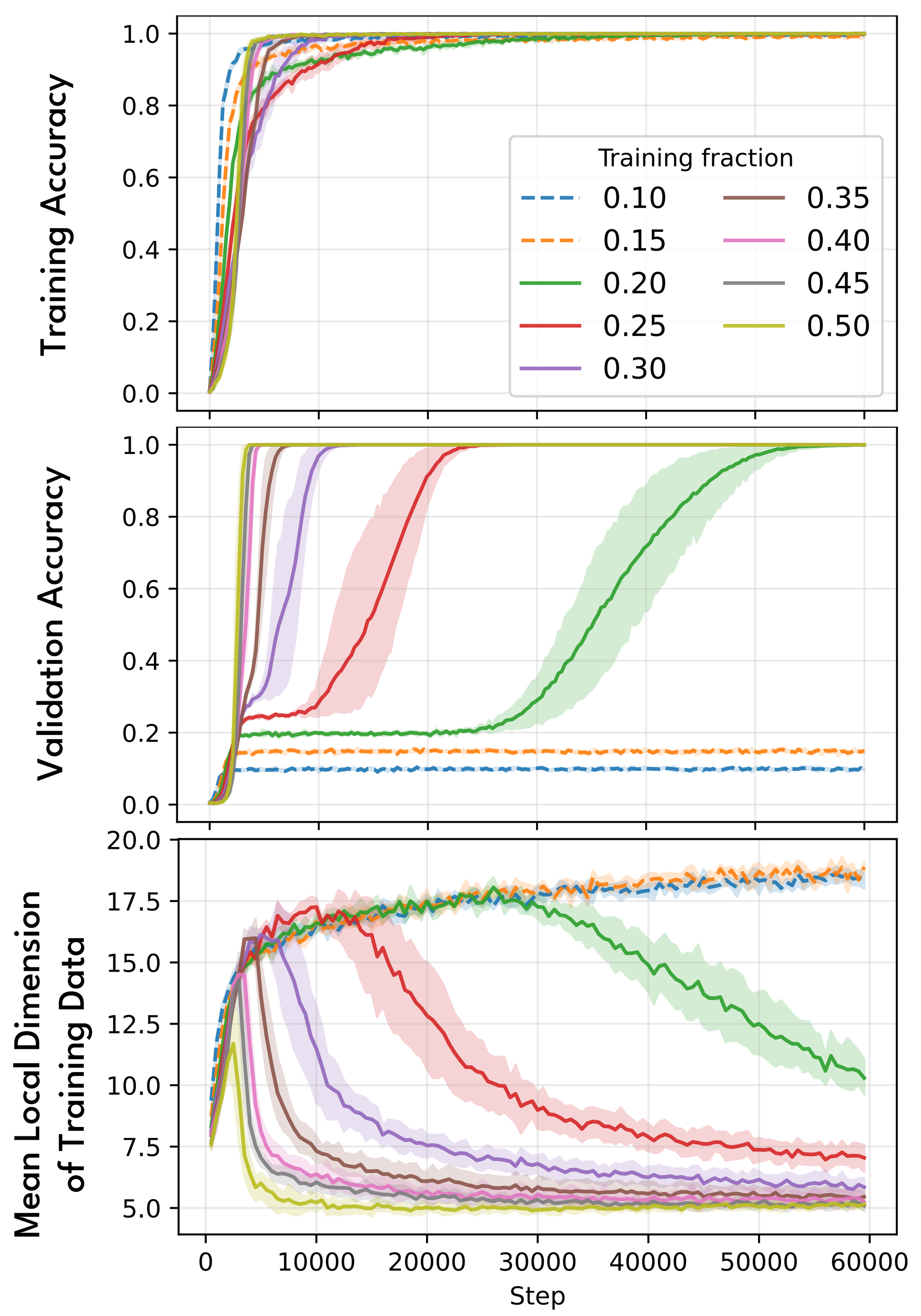


Mean dimension **drops on training data** & remains largely **unchanged on unrelated data**.

Across diverse tasks, a sustained drop in mean local dimension reliably predicts improved generalization.

Local intrinsic dimensions	Potential of geometric descriptors to
Valuable unsupervised signal for practitioners	Complement traditional evaluation methods
Usage: Interpret and monitor LLM behavior	Inform future model design

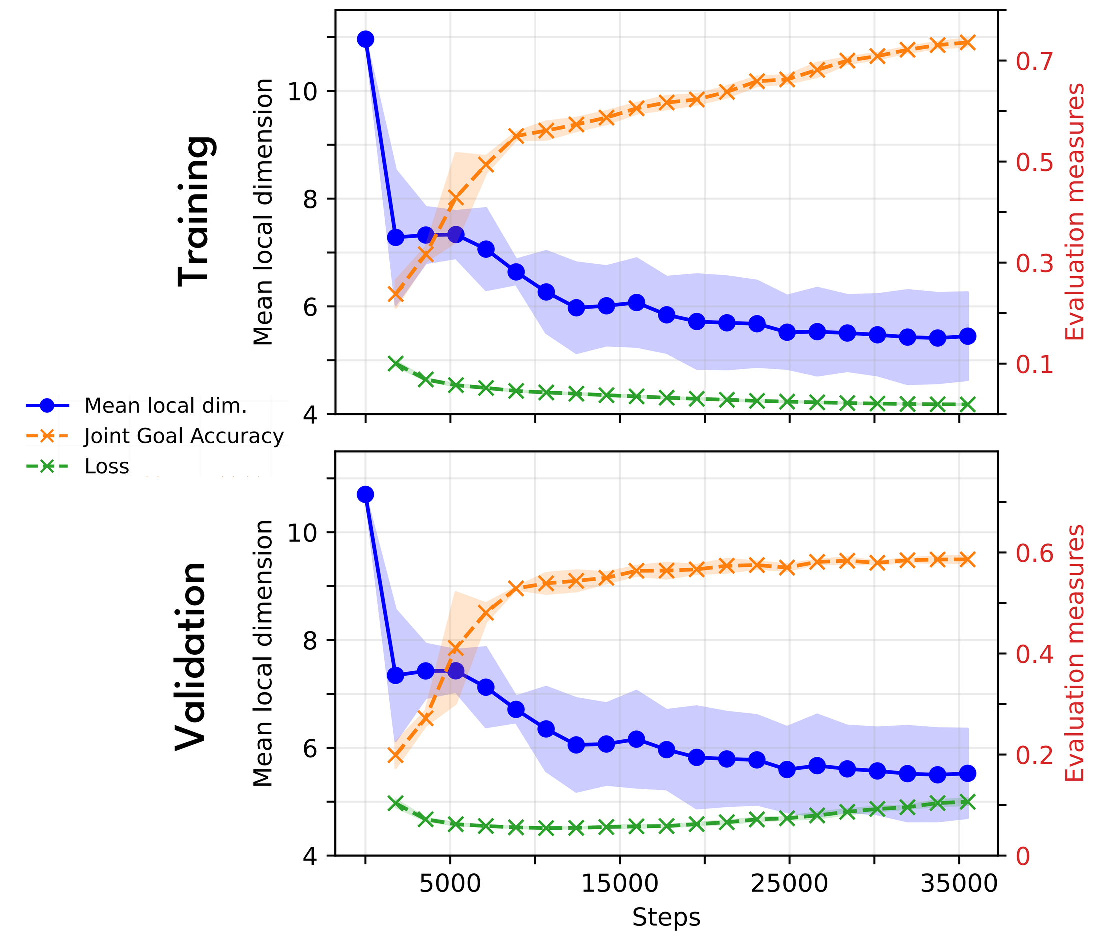
(2) Dimension Drop Anticipates Grokking



Task: Modular arithmetic operation on a small group.
Grokking: Machine learning model acquires specific skills only after extended training far beyond saturation on the training set.

Drop of embedding dimension signals the onset of **generalization** in grokking.

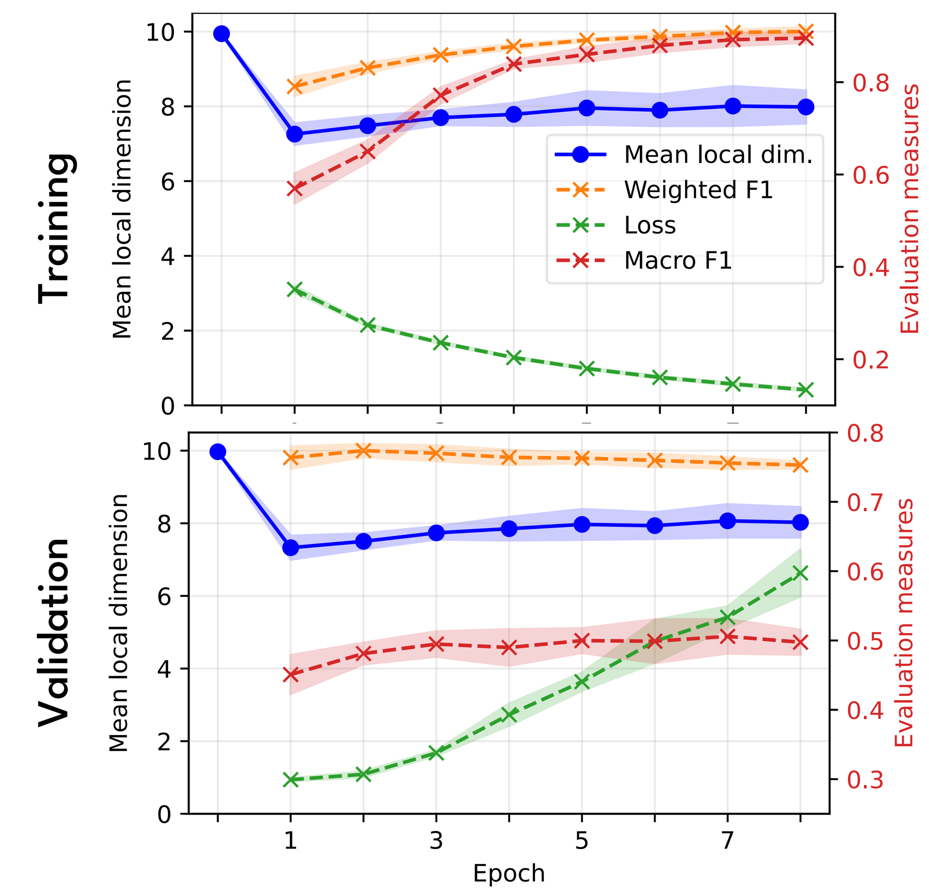
(3) Dimension Stabilization Mirrors Learning Convergence



Task: Sequence-tagging-based Dialogue State Tracking

For **training convergence**, trust **stabilizing dimension** over validation loss.

(4) Dimension Increase Detects Overfitting



Task: Classify dialogue utterances into seven emotion classes.

Initial drop followed by rise in dimension mirrors trade-off between **generalization** and **memorization**.

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